

WHAT'S THE VALUE OF ONE LIFE SAVED?

Training in the whole-house approach to home performance gives contractors the ability to save lives as well as energy.

by LARRY ZARKER

On a chilly morning in early November last year, Pete Foster arrived at the home of Hella Baer on Long Island, New York, to perform an energy evaluation of her home and to provide her with recommended upgrades. When Foster entered Ms. Baer's house, the level indicator on his CO detector immediately began to rise. It reached 18 ppm in the living room. Foster is certified as a Building Analyst by the Building Performance Institute (BPI), a national nonprofit organization that provides professional credentialing for organizations and individuals in the home performance industry. Current BPI standards state, "Diagnostic evaluations and inspections must be aborted if ambient CO concentrations greater than 35 ppm are recorded." Though CO levels at this point were not above 35 ppm, Foster was still concerned. Carbon monoxide can cause nausea, headaches, and dizziness even at levels below 10 ppm.

A Timely Intervention

Foster was sent to Ms. Baer's home through the Long Island Power Authority (LIPA) Residential Energy Affordability Partnership low-income program (REAP) (see "Protecting the Most Vulnerable"). Rather than giving the customer the usual summary of his job duties, Foster inquired about her health. Ms. Baer said that she hadn't been feeling well lately. Foster opened some windows to ventilate the home so that he could investigate the source of



MICHAEL PATREY

A healthy Hella Baer stands in front of her home on Long Island, New York.

the elevated CO level. He first checked to ensure that the furnace's induced draft fan was working properly. Another potential source of elevated CO levels is a cracked heat exchanger in the furnace. The air that is pulled through the furnace does not come into direct contact with the flame unless the heat exchanger is cracked, in which case the by-products of combustion, including CO, can enter the duct system for delivery into the home. Foster found no problems with the furnace.

Foster then investigated the gas-fired water heater. He found that an enclosure had been built around the hot water tank. In order to reach the tank, he had to remove a panel, whereupon the ambient CO level rose rapidly up to and then above 50 ppm. He immediately put the panel back in place, evacuated the customer and himself, and called Keyspan, the local gas utility,

asking them to shut off gas service until the problem could be corrected.

All combustion appliances require adequate oxygen to burn fuel efficiently. In this case, the burner for the gas-fired water heater was not receiving sufficient oxygen. Incomplete combustion was causing CO levels that were much higher than normal. In addition, the water heater tank enclosure was keeping the natural draft appliance from venting properly. CO was deposited into the living space rather than being vented to the outdoors.

Noting that Ms. Baer was looking quite ill, Foster promptly called the fire department, which dispatched an ambulance to take her to the Brookhaven Memorial Hospital Medical Center. There she was treated for carbon monoxide poisoning. Further testing done by the fire department showed that the oxygen level in Ms. Baer's home was

Protecting the Most Vulnerable

The Residential Energy Affordability Partnership (REAP) is part of the Long Island Power Authority (LIPA) Clean Energy Initiative portfolio of commercial and residential efficiency programs. REAP is designed to address the electrical-efficiency needs of LIPA customers whose household

incomes are at or below 70% of service territory median income. (Service territory consists of Nassau and Suffolk counties, as well as the Far Rockaway peninsula, on Long Island.)

LIPA is particularly concerned for the health and safety of its customers. The power authority works closely with Keyspan and first-response organizations to ensure that gas-related problems, such as gas leaks and

other combustion appliance safety issues, are identified during the REAP site visit. Honeywell Utility Solutions, the LIPA delivery contractor, employs a social worker for REAP to encourage appropriate remediation of health and safety problems through other organizations and resources, since these problems cannot be addressed within REAP.

critically low. The fire department told Foster that it was a good thing that he showed up when he did. Colder weather was expected that evening, and if Ms. Baer had tightened up the house a little bit more by not opening windows for ventilation, and if she had not gone in and out as much because of the cold, she could have died. Ms. Baer was relieved to have the ordeal behind her. When asked how long she was hospitalized, she quipped, “I ran out of that hospital as fast as I could. How many 85-year-olds do you know who go into the hospital and come back out standing?”

CO Dangers are Real

Not all CO stories have such a happy ending. In Troy, New York, on December 8, 2006, Hoan Son, a Vietnamese immigrant age 61, died in his bed from carbon monoxide poisoning. His wife, Lan Son, age 55, was rushed to Jacobi Medical Center, where she was admitted in very critical condition. Unfortunately, she died four days later. An investigation showed that the safety switch on the direct-vent kit in the furnace of the 13-year-old home was wired incorrectly. The vent consists of an electric fan that expels dangerous combustion gases through an outside vent. As a result, if the fan unit fails, the incorrectly wired safety switch does not shut off the burner, allowing the deadly gas to fill the home. A similar problem was found in neighboring homes along the same street. The homes were built through the Troy Rehabilitation and Improvement program in the early 1990s.

One week later, Stewart Townsend and his 16-year-old son, Nicholas, died from CO poisoning in Windham, Maine from fumes from a generator that they were running in their home. The Townsends were using the generator, which was in an unventilated basement, because electric service had been cut off in June. CO poisoning from a generator is blamed for two other deaths in Maine in November 2006, but those deaths had nothing to do with a power disconnection. The two men who died were working on the interior of a house under construction and died after being overcome by fumes.

Nationwide, there are over 200 deaths each year as a result of carbon monoxide poisoning. The Consumer Product Safety Commission (CPSC) estimates that 71% of these deaths occur in homes, and 55% of those are related to heating systems. Under certain conditions, all appliances that burn fuels can leak deadly CO. These fuels include kerosene, oil, coal, both natural and liquefied petroleum gas, and wood.

CPSC urges consumers to have a professional inspection of all fuel-burning heating systems—including furnaces, boilers, fireplaces, water heaters, and space heaters—to detect potentially deadly CO leaks. “Having a professional inspection of your fuel-burning heating appliances is the first line of defense against the silent killer, carbon monoxide poisoning,” said CPSC Chairman Hal Stratton. CPSC also recommends that every home have at least one CO detector that meets one of these standards: Canadian Standards Association 6.19-01, 2001; Underwriters’ Laboratories 2034, Second Edition, October

1998; or the International Approval Services 6-96, Second Edition, June 1, 1998.

Ms. Baer agrees with these recommendations and suggests that everyone install at least one CO detector in his or her home. She now has two. She also recommends that consumers hire professionals who are trained and certified to use the equipment that can diagnose health and safety problems when they are working on a home. “I was not aware that anything was wrong,” she says, “and I thank Mr. Foster for saving my life.”



Larry Zarker is the director for national development for BPI in Malta, New York.

For more information:

For more on the Building Performance Institute and certification for home performance contractors, visit www.bpi.org.

For more information about LIPA and its low-income safety and energy efficiency programs, visit www.lipower.org/residential/outreach.html.

And for more about Honeywell Utility Solutions, contact Scott Markwood
28 Main St., Building 8
North Easton, MA 02356
Tel: (973)890-9500
E-mail: scott.e.markwood@Honeywell.com